Executive Report: Olist S.A. E-commerce

Final Project– Logistics Performance Dashboard in Power BI

Date: 15/04/2026

1. Introduction and Company Context

Olist is a Brazilian e-commerce platform that connects over **100,000** small and medium-sized **sellers** with millions of customers nationwide, acting as an intermediary in order management, payments, and logistics. It operates through an integrated marketplace model that centralizes the operations of sellers distributed across Brazil's 27 states, managing the complete order cycle from multiple origins to the end customer.

In a highly competitive environment where delivery speed and reliability are critical factors for customer satisfaction, **Olist** faces the challenge of optimizing its logistics performance. Efficient management of the entire order cycle—from purchase to delivery—is key to maintaining competitiveness and improving the user experience. In this context, the company seeks to enhance its decision-making process by using data analysis tools to monitor logistics performance, detect inefficiencies, and anticipate operational problems.

2. Problem Definition and Analysis Objectives

2.1. Identified Problems

- Lack of visibility into the performance of the logistics process.
- Difficulty analyzing order delivery times.
- Absence of clear indicators regarding delivery delays.
- Limited ability to identify temporal patterns in orders.
- Lack of comparative analysis by geographic region.
- Poor monitoring of the order preparation process.

2.2. Analysis Objectives

The main objective is to develop an interactive Dashboard in Power BI, named the ***Logistics Performance Dashboard***, to:

- Analyze the logistics performance of orders.
- Measure delivery and preparation times.
- Identify delayed orders.
- Analyze temporal trends in orders.
- Evaluate differences by geographic region.
- Facilitate data-driven decision-making.
- Analyze the relationship between logistics performance and customer satisfaction.

2.3. Key Questions to Answer

- What is the **average delivery** time for orders?
- What is the **average order preparation** time?
- What **percentage** of orders is **delivered with a delay** beyond the estimated timeframe?

- Are there significant differences in **delivery times by region**?
- What **order statuses** are most frequent?
- Are there **periods** with a **higher incidence of delays**?
- Which **product categories** present the **most logistics issues**?
- Where are the **main logistics bottlenecks** located?
- How does **order volume** evolve **over time**?

Complementary Analysis *(Customer Satisfaction & Delivery Performance):*

- What is the **average rating** of orders?
- How do **delivery delays** affect **customer ratings**?
- Do **delayed orders** receive **worse reviews**?
- Which **regions** have the **lowest satisfaction**? How does **satisfaction** evolve **over time**?

3. Data Model Explanation and Applied Metrics

3.1. Data Sources

The analysis is based on the public dataset "***Brazilian E-Commerce Public Dataset by Olist***", composed of multiple **CSV files**, including:

- `olist_orders_dataset.csv` (Orders: status, purchase, approval, shipping, and delivery dates).
- `olist_customers_dataset.csv` (Customers: identifier, city, and state).
- `olist_products_dataset.csv` (Products: category, weight, and dimensions).
- `olist_order_payments_dataset.csv` (Payments: type, installments, and amount).
- `olist_sellers_dataset.csv` (Sellers: location by city and state).

- olist_order_reviews_dataset.csv (Reviews: score, comment, and dates).

3.2. Preparation and Modeling Process

a) Cleansing and Transformation (Power Query)

The following **transformations** were carried out:

- **Data type conversion** (dates, text, numeric).
- Removal of **unnecessary columns**.
- Removal of **duplicates** in dimensions.
- Handling of **null values**.
- Merging queries between the original **fact table** (olist_orders) and the **dimension tables** (using a **left outer** join) to obtain a **single Fact Table** with **all relevant information** for the analysis, thus avoiding a snowflake data model.
- **Renaming** tables (e.g., Fact_Orders, Dim_Customers).
- **Derived Columns: Key columns** were created for the analysis:
    - **Delivery Time:** Time between purchase and delivery.
    - **Processing Time:** Preparation time from order placement to carrier handover.
    - **Delivery Delay:** Difference between actual and estimated delivery.
    - **On-Time Delivery:** Classification of orders as on-time or delayed.
    - **Order Date:** Date data type column instead of date and time.

b) Data Model (Star Schema)

The **data model** was designed following a **star schema** to **optimize analysis performance** and facilitate data interpretation. A **main fact table** was built by integrating **multiple data sources**, consolidating relevant *order*, *payment*, and *valuation* information into a single structure.

- **Fact Table:** Fact_Orders (central table collecting **full order cycle information**, including *logistics*, *economic*, and *customer satisfaction* data).
- **Dimension Tables:** Dim_Customers, Dim_Products, Dim_Sellers, and Calendar (generated via DAX).
- **Relationships** were defined as **1-to-Many (*)** between **each dimension** and the **fact table**, ensuring referential integrity and facilitating multi-dimensional analysis.

c) Main Metrics (Logistics Analysis - DAX)

- **Total Orders** = DISTINCTCOUNT(Fact_Orders[order_id])
- **Average Delivery Time** = AVERAGE(Fact_Orders[DeliveryTimeDays])
- **Average Processing Time** = AVERAGE(Fact_Orders[ProcessingTimeDays])
- **Delayed Orders** = CALCULATE(COUNTROWS(Fact_Orders), Fact_Orders[DeliveryDelayDays] > 0)
- **% Delayed Orders** = DIVIDE([Delayed Orders], [Total Orders])
- **% On-Time Delivery** = 1 - [% Delayed Orders]
- **Max Delivery Time** = MAX(Fact_Orders[DeliveryTimeDays])

d) Complementary Metrics (Customer Satisfaction)

- **Avg Review Score** = AVERAGE(Fact_Orders[review_score])
- **Total Reviews** = COUNT(Fact_Orders[review_id])
- **% Positive Reviews** = DIVIDE(CALCULATE(COUNTROWS(Fact_Orders), Fact_Orders[review_score] >= 4), [Total Reviews])
- **Avg Score Delayed** = CALCULATE([Avg Review Score], Fact_Orders[DeliveryDelayDays] > 0)

4. Dashboard Design and Development

The dashboard adopts a **progressive** analytical model, divided into three main tabs that guide the user from the **macro vision** to the **customer impact**, relying on pop-up detail pages (**tooltips**) to avoid visual clutter.

- **Tab 1: Logistics Overview:** Global KPIs, temporal analysis (line chart), operational status (adjusted horizontal bar chart), and volumetric geographic analysis (point map).
- **Tab 2: Operational Analysis (Bottlenecks):** Operational KPIs, Performance Matrix by Product Category (scatter plot crossing delivery time and delays), and Bottleneck Ranking (Seller matrix and regional bar chart with data bars).
- **Tab 3: Customer Satisfaction:** Quality KPIs, Satisfaction distribution (column chart 1-5 stars), Cross-evolution (combo chart comparing volume vs. score), Logistics Performance Impact (100% stacked bar chart contrasting on-time vs. delayed scores), and Satisfaction Map (Choropleth).
- **Hidden Detail Levels (Interactive Tooltips):**
    - ***Geographic Tooltip*** *(Tab 1):* Hovering over a **state** reveals **revenue**, **average review score**, and **Top 10 categories**.
    - ***Product Category*** *Tooltip (Tab 2):* Hovering over a **category** reveals **total orders**, **average delivery time**, and an **on-time vs. delayed** pie chart.

4.2. Visualizations and Usability

To ensure an **agile** analytical experience:

- **Dynamic KPI cards** for immediate reading.
- **Advanced charts** (scatter plots, combo charts).
- **Conditional formatting** in matrices and bar charts to quickly identify **top offenders**.
- Precise **interactive maps** (point and choropleth).
- **Pop-up pages** (Tooltips) for extra depth without clicks.
- **Global slicers** (Date, Region, Status, Category).

- **Design principles:** Clean layout, coherent color scheme (green/red alerts), and total interconnectivity (cross-filtering).

5. Key Analysis Insights

Logistics Performance:

- The **average delivery time** across the dataset is around **12 days**, with significant variability by destination state.
- Approximately **6,57%** of orders accumulate **delays against the estimated date**, with extreme cases exceeding 30 days.
- The **average preparation time** is **3.2 days**, identified as the **main bottleneck** of the logistics cycle.
- While **delayed orders** represent a minority, their **impact** is **disproportionate**: the average **rating** of **on-time orders** is **notably higher** (4.3/5) compared **to delayed orders** (2.5/5).

Temporal Trends:

- Order volume shows **sustained growth** (2016-2018), with peaks in **November** (Black Friday) and a **generalized increase** in the **second half** of the year.
- These **high-demand periods** correlate with a **slight increase in delivery times** and the highest percentages of **delayed orders** (November and January), suggesting a **lack** of scalable **logistics capacity**.

Geographic Analysis:

- **Northern** and **northeastern states** (e.g., Pará, Amazonas) record delivery times well above the national average, while **São Paulo** and **Rio de Janeiro** are consistently **below**. This disparity is tied to **carrier density** and **distance to major hubs**.
- Regions with the **highest delay rates** also coincide with the **lowest average satisfaction** scores.

Preparation Process:

- **Preparation time** represents, on average, between **20% and 25%** of **the total delivery time**. In **delayed orders**, this percentage is **significantly higher**, confirming it as the area with the **greatest improvement potential**.

Customer Satisfaction:

- Global **average score** is **4.1** /5. However, **delayed orders** plunge to **2.5**.
- **77% of reviews** are **positive** (**4-5 stars**), whereas **delayed orders** concentrate the **highest percentage** of **negative reviews** (1-2 stars), evidencing the **direct impact** of logistics on **platform reputation**.

6. Strategic Recommendations

1. **Optimize the Preparation Process:** Establish **internal SLAs** for **maximum preparation time** and **automate alerts**. Reducing **processing time** is the lever with the highest impact on customer experience.
2. **Proactive Delay Management:** Implement an **early warning system** for orders at **risk of delay** before deadlines are missed. Prioritize **proactive communication** with the customer.

3. **Seasonal Operational Planning:** Anticipate the **reinforcement** of logistics resources during **identified high-demand months** (November, January), negotiating capacity with carriers well in advance.
4. **Logistics Improvement in Critical Regions:** Design **specific action plans** for northern and northeastern states. Explore **agreements** with **local operators** and **intermediate distribution hubs** to reduce last-mile distances.
5. **Continuous Monitoring:** Establish a **periodic review routine** using the ***Logistics Performance Dashboard***, assigning owners to KPIs and setting **predefined alert** thresholds.
6. **Reduce Delays as a Satisfaction Lever:** Since **delayed orders** receive **ratings 40% lower** than on-time ones, **reducing this metric** yields the **highest expected return** in loyalty and reputation.
7. **Incorporate Reviews as a Strategic KPI:** Integrate **average score** and **positive review %** as **top-tier indicators** in operational tracking.

7. Conclusion

The development of the ***Logistics Performance Dashboard*** provides **Olist** with an **effective tool** for monitoring and analyzing **logistics performance**. Thanks to a structured data model, key indicators, and interactive visualizations, the dashboard facilitates **data-driven decision-making**, allowing the **identification** of **inefficiencies**, **process optimization**, and an **enhanced customer experience**.

Adopting this tool positions **Olist** to **anticipate** logistics problems, **act** with greater agility, and **build** a sustainable **competitive advantage** based on **service reliability**.